Design and Implementation of High-Throughput 175-Tap FIR Filter Architectures

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Abstract

This report presents a comprehensive design space exploration of 175-tap FIR filter architectures. By analyzing performance metrics such as area (ALMs), power, and F_{max} for seven distinct designs, we demonstrate the throughput advantages of parallel and pipelined structures.

1 Introduction

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2 MATLAB Design Methodology

2.1 Filter Specification

- **Type:** Low-Pass Equiripple (Parks-McClellan)
- **Taps:** 175
- **Transition Band:** 0.2π to 0.23π rad/sample
- **Stopband Attenuation:** > 80 dB

2.2 Quantization and Fixed-Point Analysis

3 Implemented Architectures

We categorize our seven implementations into four architectural paradigms:

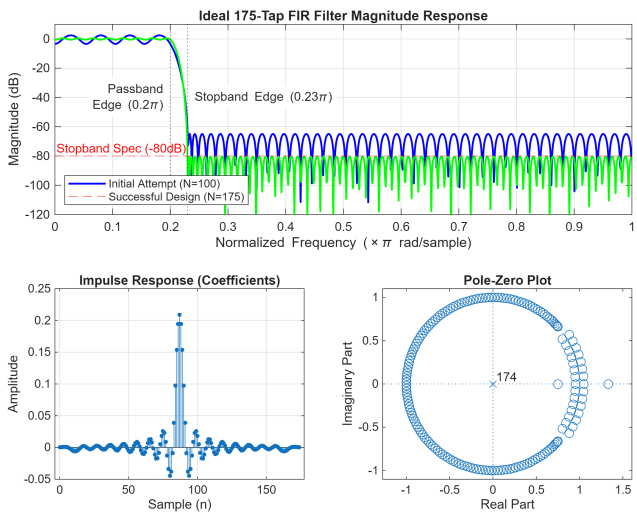

Figure 1: MATLAB frequency response: Original vs. Quantized.

3.1 Baseline and Pipelined Designs

3.2 Standard Parallel Architectures

3.3 Reduced-Complexity Parallel (Fast FIR)

3.4 Pipelined Fast FIR (Optimal)

4 Synthesis Results

All designs were synthesized and fitted for the Intel Cyclone V FPGA (5CGXFC7C7F23C8) using Quartus Prime Lite.

Table 1: Hardware Performance Comparison

Architecture	**ALMs**	**DSP**	F_{max} **(MHz)**	**Power (mW)**
Direct L=1	2,003	156	34.1	122.56
Pipelined L=1	2,016	131	68.9	93.02
L=2 Parallel	14,119	156	36.8	227.98
L=3 Parallel	24,726	156	35.4	354.13
L=2 Fast FIR	18,785	0	35.4	201.87
L=3 Fast FIR	25,604	0	35.0	302.06
Piped FFA L3	**14,270**	**156**	**68.4**	**137.37**

5 Performance Analysis

6 Functional Verification

RTL verification shows bit-exact matches with MATLAB models across all stimuli.

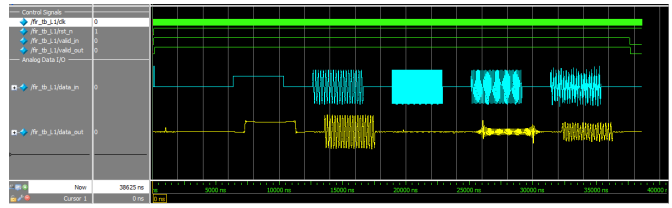

Figure 2: Functional verification: Waveform for [Arch Name].

7 Conclusion

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References